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The new role of data in society: Challenges and opportunities for mathematics education in the age of data science and AI

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Data Science and Artificial Intelligence (AI) are reshaping society, entangling (Coles & Hiis Hauge, 2025) mathematics with people, values, and the wider world. This transformation brings new challenges and opportunities for mathematics education, while also calling for broader curricular shifts that connect with other school subjects. In this talk, we examine key skills and literacies students need to navigate a data-driven world and imagine their role in shaping it—critical data literacy, statistical, media, computational, and AI literacy. We discuss how these literacies should evolve within mathematics education and emphasize the importance of interdisciplinary collaboration across data science, the natural and social sciences, and computer science. Classroom examples from our own projects illustrate how these ideas can be brought into practice.

Keywords: Data science education, Data literacies, AI literacy.

1. Introduction

The topic of data has always fallen between the cracks—especially between mathematics and computer science:

> Computer scientists, by nature, don't respect data. They have traditionally been taught that the algorithm was the thing, and that data was just meat to be passed through a sausage grinder.
>
> So to qualify as an effective data scientist, you must first learn to think like a real scientist. Real scientists strive to understand the natural world, which is a complicated and messy place. By contrast, computer scientists tend to build their own clean and organized virtual worlds and live comfortably within them. Scientists obsess about discovering things, while computer scientists invent rather than discover. (Skiena, 2017, p. 2)

We invite readers to consider whether there might be a similar attitude toward data among mathematics educators—that the term "computer scientists" in the passage above might be replaced with the word "mathematicians". Of course, data are part of statistics education, which is taught in the mathematics classroom, and data play a certain role in mathematical modelling. In computer science education "data and information" is a standard topic. And as the quote highlights, data are also central to inquiry across the natural and social sciences.

However, standard current educational practice is far away from taking the emergent new roles of data in society into account. Classes and curricula are still typically organized according to relatively siloed and static visions of school subjects, limiting the extent to which scientific, social, mathematical, and computational explorations can be meaningfully brought together to enable students to make discoveries with data. At the same time, the confluence of all these disciplines and approaches makes planning developmentally appropriate and tractable activities with data

challenging. Exactly what should students know about data, and when should they learn it? And, how can educators be prepared to teach such a dynamic and interdisciplinary set of practices?

In this paper, we review the current state of the field of data science education and provide some examples from our own research and development work. It is based on a plenary talk we gave at CERME 14 in February 2025, where the slides contain many illustrating examples (https://bit.ly/cerme14). We begin by defining *data science* as an emerging, interdisciplinary field and examining its impacts on society, science, and education. We then present pedagogical examples of data science driven activities, drawn from our own work and others at the interface with the natural, social, and computer sciences. Our goal is to plot a course for considering the impact of data science, and related innovations such as machine learning and generative AI, *across* the curriculum—and the implications of these curricular impacts for mathematics education. We invite our readers to consider, within this context:

- What is important for students to know about data?
- What should students be able to do with data?
- What is (or should be) the role of mathematics education in contributing to the knowledge, skills, and dispositions needed for students to understand life with data?

A first answer, which one could have provided at least 20 years ago is: Students should have experience with analysing "tidy data" (Wickham, 2014) with digital tools from the perspective of statistical investigation (Wild & Pfannkuch, 1999). Tidy data are multivariate data, organized in a table in rows and columns, like a survey where each row is a person and columns record their attributes. Attributes can be numerical or categorical. Digital tools for school use have been available (Biehler et al., 2013), as well as material, data sets, and ideas for classroom use, backed by research (see e.g., Ben-Zvi et al., 2018, and the website of the International Association of Statistics Education iase-web.org). However, implementation in standard curricula is far behind the consented goals among statistics educators in many countries.

The new developments of data in science and society demand an even more substantial place of data in the curriculum, which can build on traditional approaches and integrate new aspects. We see the emerging societal pressure on schools to integrate more data literacy or AI literacy as helpful for curricular reform. The Organization for Economic Co-operation and Development (OECD), for example, recently announced an effort to develop an AI literacy framework that extends across the curriculum, so that "educators will be able to better assess students' understanding of AI, as well as broader issues like privacy concerns and digital citizenship with clear, measurable benchmarks [...] not only for academic success but also for active, well-informed participation in an AI-driven, digitally evolving society" (Schleicher, 2025).

2. Characterizing Data Science and its Evolving Meanings and Roles in Society

What counts as data has changed dramatically and rapidly. Data now includes everything from images, text, videos, and webpages to audio data (speech), sensor data, and records of transactions. We have to think also about posts to social media, location data from GPS or location-based services, and even behavioural or health data collected by wearable devices. The explosion of digital data means that conceptions of data literacy, the ability to access, interpret, critically assess, manage, handle, and ethically use data, has to be updated as well as what we understand as democratic participation in a society being transformed by datafication.

Preparing K-12 students for this data-rich reality means ensuring they understand not just how to calculate with numbers, but how to make sense of, use, and critically evaluate data in its many forms.

This changing nature and increasing availability of data have led to new and significant roles for data, data-based models, and data-related algorithms across various spheres.

Data play a new and expanding role in the sciences. In fields like statistics and mathematical modelling, we now encounter new types of data and require new workflows. For instance, the CRISP-DM workflow model (Shearer, 2000) provides new emphasis on data preparation, communication, and deployment. It differs from the traditional PPDAC cycle (Wild & Pfannkuch, 1999, discussed in more detail in Section 4.1) and mathematical modelling cycles, even if they include data as a separate component (English, 2021). Related to the new role of data, new types of models become relevant: predictive models (the dominant type of model in machine learning), algorithmic and computational models (Breiman, 2001), sometimes with millions of parameters. Computer science has seen the emergence of data-driven Artificial Intelligence (AI) and data-driven digital artifacts. Even in natural, social, and life sciences, datafication and new research methods are transforming how questions are asked and answered. For example, data science techniques are being used to understand complex biological systems.

At the same time, data science is emerging as a new discipline itself. It is an interdisciplinary field that combines statistical methods, computer science, and expertise in a particular subject area to gain insights from data. Hazzan and Mike (2023, pp.19) devote a whole chapter to the origins and different characterizations of data science.

Perhaps most visibly for young people, data has new roles in everyday life, including the lives of students. Students interact constantly with data-driven digital artifacts in their private lives through search engines, streaming services, and social media. Wearables and fitness trackers collect personal data. Digital data are woven into the fabric of our cultural, social, political, and economic worlds. Educational conceptions that focus on these aspects of data can be found under the heading of personal data literacies (Pangrazio & Selwyn, 2019) and building students' data awareness (Höper & Schulte, 2024).

Similarly, data and its visualizations have new roles in the media. We see more data and new visualizations in traditional media, as well as websites with embedded interactive visualizations and dashboards. Data journalism is an emerging field. Understanding data in the media requires a new lens, especially when dealing with spatial data or issues of spatial justice (da Silva et al., 2021; Engledowl & Weiland, 2021; Rubel et al. 2021). Media items use data and visualizations to inform the public, present opinions, justify predictions or recommendations, and validate government policies. Data visualizations are not neutral; they are social texts authored from specific points of view that narrate particular stories (Gal, 2022).

Beyond consumer applications, data-based models, and data-related algorithms have profound new roles in society at large. Algorithmic decision systems, often based on machine learning, are used in healthcare (predictive diagnosis, treatment recommendations), finances, transportation, job applications, educational administration, and predictive policing. Many applications raise ethical questions. Examples include examining racial bias in healthcare AI (Chin et al., 2023), predicting school dropouts (Bowers, 2021), or the controversial use of tools like COMPAS software in the justice system (https://en.wikipedia.org/wiki/COMPAS_(software)).

These new roles of data have led to significant political discourses on the ethical aspects of data and AI use, involving associations, initiatives, and state regulations. New scientific journals are devoted to this topic such *AI and Society* and *Big Data & Society*. Key concerns raised in the 400 pages report of Deutscher Ethikrat (2023) include the danger to individuals through statistical stratification, the protection of privacy and autonomy versus threats from surveillance, and the pervasive issues of bias

and discrimination embedded in data and algorithms. Transparency and traceability are crucial for control and responsibility. Counterbalancing these concerns are efforts focused on AI and Data Science for Social Good (https://www.datascienceforsocialgood.org). Critically evaluating data use also requires examining data quality and the strength of evidence presented, issues that are increasingly discussed in the media.

Given this landscape, preparing students for a data-driven world involves fostering data literacy that includes the ability to critically interpret and evaluate data and its sources. This is reflected in educational conceptions such as *critical data literacy* (Weiland, 2017). Data visualizations, while powerful tools for communication, are embedded within structural and ideological relationships of power and can become sites where meanings are contested. Access to learning about data visualization without a deep interrogation of identity and status, national and commercial interests, and power relations can even be counterproductive, potentially reproducing societal hierarchies in the classroom (Bilstrup et al., 2022; d'Ignazio & Klein, 2020; Hall & Dávila, 2022; Lee et al, 2022; Philip et al., 2016). Students need to understand that data can be used to both uphold and expose systems of oppression (Martínez-Castro et al., 2023). Critically reading data visualizations involves understanding how reality is mathematically formatted, how variables and relationships are framed, and what stories are narrated, recognizing that these are texts created by humans with specific viewpoints. Ignoring the social complexities of data and its representation would defeat the democratic purpose of teaching data literacy.

3. How is the Growing Role of Data in Society Impacting K-12 Policy?

Much has been written about the role of data in educational administration and decision-making (e.g., Mandinach et al., 2015; Schildkamp & Datnow, 2022). While these impacts of data are certainly important, here we are interested in a complementary issue: what must students themselves now learn, given the increasing role of data in science and society?

To address this question, we consider what Chevallard called *didactic transposition* of knowledge (1985). Didactic transposition refers to the process by which scholarly knowledge is transformed to become teachable. The challenge with data science is that the relevant scholarly knowledge is in itself interdisciplinary and unstable, and practitioners have different perspectives regarding its nature and function as a field (Tanweer & Steinhoff, 2024). This makes it difficult to identify a stable body of knowledge to be taught or locate where data science might live within the broader curriculum. Chevallard also notes that didactic transposition can introduce tensions between maintaining the legitimacy of the taught knowledge in the eyes of experts and adapting that knowledge for novices in ways that are accessible and socially relevant. This tension is already playing out as school systems grapple with whether and how to introduce data (see Boaler et al., 2024 for a recent example), and how to prepare students for its significant impacts on society outside of academic disciplines, as described in the prior section.

One approach educators have taken to address these tensions when introducing data science at the K-12 level is through integration with a variety of existing school subjects. A recent position statement authored jointly by a collection of organizations in the United States representing teachers of Mathematics, Statistics, Science, Social Studies, and Computer Science argued that "Data science bridges disciplines and thus should be introduced and taught across the curriculum in K-12 schools [...] All subjects in school should recognize the contribution of data to their discipline and take curricular approaches that integrate data with disciplinary lessons where appropriate" (NCTM et al., 2024, p. 1). Such initiatives require a developmental approach to coordinate data science integration

across the grades and subjects, leading some organizations to call for the development of comprehensive frameworks to inform curriculum and instruction (DS4E, 2025; NASEM, 2023).

Cross-curricular integration can be challenging, however, as it requires coordination across different academic departments and potentially changes to existing curricula. To address these challenges, integrative approaches typically emphasize that data science is fundamentally an *investigative* process that involves formulating a question, collecting or considering data, analysing the data, and interpreting and communicating results (Lee, Mojica, et al., 2022). Such references to this cycle of investigation are increasingly emphasized in new related standards and frameworks (e.g., Bargagliotti, 2020; IDSSP Curriculum Team, 2019). And while they may use different terminology, the core concept of inquiry—pursuing a compelling question or evaluating a claim using data—is shared across many disciplines (Manz & Beckert, 2023; Schanzer et al., 2022; Shreiner, 2018; Wild & Pfannkuch, 1999).

Another approach educators have taken to introducing data science at the K-12 level has been through its connections to citizenship education. This approach positions data as a way to empower students to understand and address real-world issues, potentially leading to pro-civic actions and understandings (Engel & Ridgway, 2022; Louie et al., 2021; Rubel et al., 2016). However, defining *what kind of citizenship* data science education aims to foster is itself a challenge. Various projects and curricula cite goals that vary from empowering individual students as (future) citizens; fostering cultural coherence and understanding across diverse societies; advancing society through supporting social mobility and equal opportunity; and boosting economic growth for students through workforce education. However, different stakeholders may prioritize these goals differently, leading to tensions in curriculum design and implementation. An emphasis on workforce preparation might focus heavily on technical skills for specific industries, while an emphasis on equal opportunity might prioritize critical data literacy for evaluating societal claims and advocating for justice (see Rosenberg & Jones, 2024). Navigating these competing priorities and ensuring that data science education effectively contributes to a balanced set of educational goals is a significant challenge for K-12 systems.

Whether data are introduced to students as a tool of inquiry across domains, or as a tool for civic and social engagement, the societal and ethical dimensions of data and its use have become increasingly important (Baumer et al., 2022; European Commission. Directorate General for Education, Youth, Sport and Culture., 2022; Utts, 2021). As a result, many of the emerging standards and frameworks described above emphasize not only data-related *knowledge and skills* but also related *dispositions* and *ethical considerations* as key learning goals for students. For example, the Dana Center's Data Science Course Framework (2024) describes design principles for courses that will "empower students to live skillfully and productively in our data-driven society" (p.4), which includes activities such as "Provid[ing] authentic experiences that empower students to make a positive difference in their communities through data science projects" (p.7).

4. How Could the Growing Role of Data in Society Impact Education and especially the K-12 Curriculum?

While administrators and policymakers are still working to understand how data might best be integrated into K-12 curriculum and instruction, many teachers, researchers, and curriculum developers have already explored new pedagogies, tools, and dispositions to introduce students to data science. A number of pedagogical frameworks concerned with data have been introduced, though typically with a focus on tertiary and professional education rather than the developmental steps needed to prepare students at primary and secondary levels for these intensive tertiary experiences. Here, we review four such adaptations and innovations related to how students are

introduced to data; what are the tools and technologies they use to analyse them; how they are encouraged to develop critical data literacies regarding how data can be used to understand and advocate for change in the world; and how is it possible to develop students' awareness of how datafication is impacting both personal life and societal structures.

4.1 New Models of the Data and Statistical Workflow

Integrating data science into K-12 education demands a fundamental rethinking of how we teach students to work with real-world quantities. For decades, reform-oriented statistics, science, and mathematics educators have emphasized the importance of moving away from solely demonstrative or obviously predetermined "cookbook" data analysis activities toward more authentic engagement in inquiry (Cobb, 1999; Hancock et al., 1992; Lehrer & and Romberg, 1996). These sorts of inquiry-based engagements with data were traditionally described as a structured cycle in which students pose their own investigative questions, collect their own data, and explore those data using appropriate tools and methods. For example, Wild and Pfannkuch's PPDAC (Problem, Plan, Data, Analysis, Conclusions; Wild & Pfannkuch, 1999) cycle is foundational to reform-oriented statistics education. However, data science, with its emphasis on new forms of data and tools, disrupts several components of the investigation cycle, requiring us to further adapt this foundation.

Perhaps the most significant shift that has been introduced to these traditional descriptions has to do with the nature and origin of the *Data* at the centre of the cycle. K-12 statistics and science lessons oriented toward student inquiry often engage students in designing experiments or conducting fieldwork in their schoolyards or neighbourhoods to collect their own small first-hand datasets. Data science, however, frequently involves working with large, complex datasets that already exist – data that are publicly "open" (Zapata-Cardona, 2025), "repurposed" (Wilkerson & Laina, 2018), and so "messy" (Rosenberg et al., 2020) that students may find themselves "awash" (Erickson, 2025).

Consider an example drawn from McDonald's *Handbook of Biological Statistics* (McDonald, 2014, pp. 204–205). Students are invited to address the question "Does species diversity decrease as you move away from the equator?" using the Audubon Society's Christmas Bird Count, a yearly participatory citizen science activity that reports the total count of species spotted in different locations around the United States. The activity presents a small subset of species counts from approximately 15 locations, along with those locations' names and latitudes. The reader is expected to run a linear regression with latitude as the independent variable and species count as the dependent variable, which yields an equation in which the species count declines slightly as latitude increases, and the result does not reach statistical significance. Inviting the student to engage in common-sense reasoning, McDonald writes, "Even if it were significant, I don't know what you'd do with the equation; I suppose you could extrapolate and use it to predict that above the 49th parallel, there would be fewer than zero bird species" (p. 205).

This activity engages students with data in meaningful and careful ways. It encourages them to consider the context and implications of mathematical methods and points out their limitations. But from a data science perspective, there is much more that can be done due to the availability and richness of data we have about the world. Consider, for example, that with the information provided (location names and latitudes), it is relatively easy to link these records with additional geographic information: with not only latitude but also longitude, daily weather patterns, elevation, geological features, and more. Merging the available bird species counts with this richer set of available information (Figure 1) reveals that the subset of bird count records selected by the author are drawn from the U. S. mid-Atlantic region, a location and scale for which bird species diversity is less likely to vary by latitude as the question might suggest, but more likely to vary by proximity to the coast.

Location	Latitude	Longitude*	Number of Species
Bombay Hook, DE	39.22	-75.48	128
Cape Henlopen, DE	38.80	-75.10	137
Middletown, DE	39.47	-75.74	108
Milford, DE	38.96	-75.46	118
Rehoboth, DE	38.60	-75.10	135
Seaford-Nanticoke, DE	38.58	-75.62	94
Wilmington, DE	39.73	-75.56	113
Crisfield, MD	38.03	-75.87	118
Denton, MD	38.90	-75.87	96
Elkton, MD	39.53	-75.86	98
Lower Kent County, MD	39.13	-76.08	121
Ocean City, MD	38.32	-75.11	152
Salisbury, MD	38.33	-75.63	108
S. Dorchester County, MD	38.37	-76.11	118
Cape Charles, VA	37.20	-76.03	157
Chincoteague, VA	37.97	-75.41	125
Wachapreague, VA	37.67	-75.69	114

* Obtained by finding locations using Google Maps.

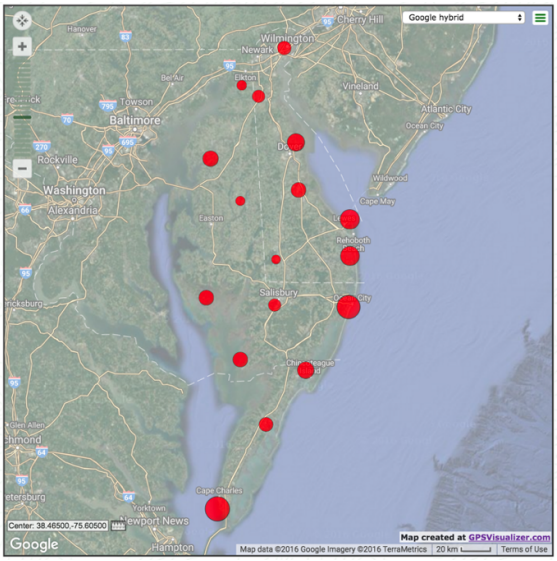


Figure 1. An expanded dataset, constructed by merging locations and latitudes with available longitude information (left) and the resulting map of bird species counts (right)

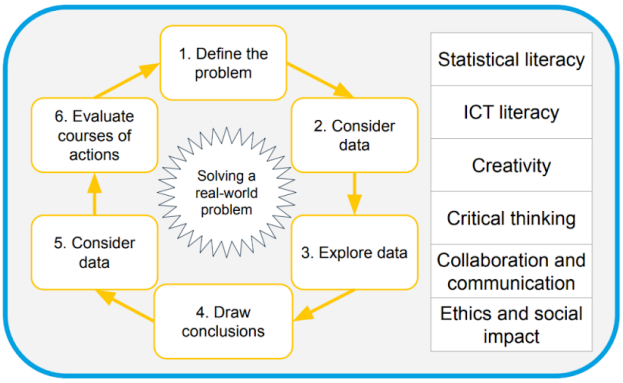
In this way, data that are collected by or provided to students can be merged, supplemented, and otherwise transformed using "found data" in ways that fundamentally change other components of the PPDAC cycle (see, e.g. Biehler & Schulte, 2018). When considering the *Planning* of an investigation, students' attention may shift from a focus on designing data collection protocols toward data preparation and exploration, emphasizing concepts of metadata, and data quality as students consider how datasets acquired from third parties likely include potential anomalies that require attention and further processing. The possibilities and constraints of an expanded, but potentially less reliable, set of data sources can in turn impact the *Problems* that an investigator is willing to take on. Instead of a linear or progressively iterative process where the problem is defined first and then data are collected, students might take a more exploratory approach with a large dataset to identify interesting patterns or generate questions.

The *Analysis* stage that results from an expanded set of available data and data types is also transformed. While traditional statistical tests are still relevant, data science more heavily leverages computational tools for exploration and visualization. As machine learning algorithms and other computational techniques become more common, investigators need to understand which data were used to train a given model and whether such training is appropriate for the new application context. To build on the bird count example, a biostatistician may wonder whether models developed from data in other geographic regions with different characteristics are appropriate to apply in this case.

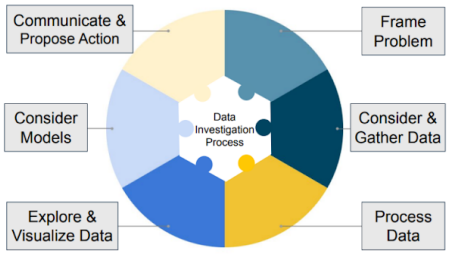
Finally, the diversity of potential data sources and analysis methods and the complex considerations each brings changes how investigators should *Communicate* the results of their analysis. They should report potential biases in data sources; shifts in paths of inquiry as a result of available or constrained datasets; evaluation of the appropriateness of the algorithms used and their training, and so on. throughout the entire process. This task has become so important that topics such as "data storytelling" have become a focus in disciplines such as the information sciences (Segel & Heer, 2010) and journalism (Matei & Hunter, 2021). Investigators are also increasingly expected to report on the ethical implications of their findings. The goal is not just to analyze data, but to use it responsibly to understand and potentially impact the world. In data science practice, communication

should take into account the principles of reproducible research (Dogucu & and Çetinkaya-Rundel, 2022). How this can be made accessible to secondary students is an open question.

In response to the perturbations to the data and statistical workflow caused by these new data sources and data types, several proposals have emerged to re-envision the data investigation cycle (Figure 2). Kazak and colleages' (2023) Data Analytics Cycle, for example, introduces two separate steps both dedicated to *considering* and then *exploring* data. The explicit inclusion of "considering" data here highlights how the increasing availability and complexity of data requires explicit attention to "getting to know" the data—including reviewing metadata, exploring the history and initial purpose for data collection, evaluating validity, and so on. Lee and colleagues' (2022) Data Investigation Cycle isolates a different dimension, data processing or the preparation, cleaning, re-coding, and otherwise transforming the data in order to permit the desired analyses, as a specific component. Wise's (2020) continuum of "data scientist perspectives" toward data further highlights data generation and storage as concerns in themselves (Figure 2).



Data Analytics Cycle & Competency Areas
Reproduced from Kazak, et al (2023)



Data Investigation Cycle
Reproduced from Lee, et al (2023)

Subject	Data Scientist					Audience
	Generation	Storage	Transformation	Interpretation	Representation	

Subject, Audience, and Data Scientist Perspectives Toward Data
Reproduced from Wise (2020)

Figure 2. New proposals for data-science-ready investigation cycles as described in Kazak et al (2023, top left), Lee et al (2023, top right), and Wise (2020, bottom)

In data practice, especially those that are influenced by computer science, even more different workflow models are used, such as the CRISP-DM cycle (Biehler & Schulte, 2018; Shearer, 2000). One major difference is that an extra stage *deployment* is added, similar to what computer science workflow models already include. Data science does not just end with a conclusion, but a system is constructed (such as a machine learning model) that is implemented in a socio-technical system, where ethical considerations have to be taken into account. Though their emphases are different, all of these proposals highlight how expanding data types and access have shifted definitions of what it means to pursue inquiry with data.

We note that while many of these frameworks are referenced by educators and researchers concerned with K-12 education, most were developed for, and/or heavily informed by, tertiary and professional levels. Even frameworks well known to the K-12 statistics and data science education community such as PPDAC (Wild & Pfannkuch, 1999) and Lee and colleagues' (Lee et al., 2022) more recent data investigations cycle were developed through interviews with professionals and statistics-oriented

students. These shed much needed light on some aspirational *goals* for modern statistics and data science education, but there is still much to be done in order to understand the *developmental processes* that might lead students to these endpoints, and to identify the available pathways through the existing curriculum that might help set the foundations for students' competence and confidence with data.

4.2 New Mathematical and Technological Tools for Working with Data

Access to new tools and technologies is also fundamentally reshaping how we should teach the data and statistical workflow in K-12 education. For decades educators have highlighted the potential for digital tools to transform students' engagement with data and modelling (Ben-Zvi, 2000; Biehler, 1997; Edelson & Gordin, 1998; Hancock et al., 1992; Lehrer & and Romberg, 1996). Topics such as probability and statistics were well situated to benefit from the introduction of computers into schools, because they allowed the easy visualization of large, multivariate datasets and the simulation of probabilistic events that form the foundations of statistical models. Ben-Zvi (2001), building from Pea (1985), argued that computers have redefined statistics as a cultural and disciplinary practice by reorganizing which statistical objects students are able to explore and manipulate, offering them fundamentally new cognitive models and new ways to connect statistical practice to the real world. Some of these same redefinitions are the ones that transformed the professional practice of data analysis and led to the emergence of data science (Biehler, 1993; Cleveland, 2001; Kaplan, 2007).

Since these beginnings, a variety of genres of educational technologies focused on data and statistical processes have become available. Several reviews and classifications of these tools exist, organized by function, interface type, or other such principles (e.g., Ben-Zvi, 2000; Biehler, 2019; Chance et al., 2007; Foley et al., 2024; Israel-Fishelson et al., 2025; A. McNamara, 2019; Pimentel et al., 2022; Pratt et al., 2011; Rubin, 2007), and we refer the reader to those for a comprehensive overview. Here it suffices to note that in general, these reviews include references to tools and technologies for collecting data from the world through automated measurements and recordings; tools for visualizing data through the easy creation of charts and graphs; tools for manipulating and calculating data; mapping tools that associate data with specific geospatial locations; and tools for generating simulated data from user-defined models. More recently, such reviews and frameworks have also expanded to consider how the design of particular tools may facilitate students' transition to professional practice (Foley et al., 2024; A. McNamara, 2019). One family of tools deserves special note here because of their impacts on statistics and data education research and because we both use them extensively in our own work. These include *Fathom* (a powerful GUI-driven educational data analysis and visualization tool; Finzer, 2001), *Tinkerplots* (a version developed for younger students, www.tinkerplots.com), and the *Common Online Data Analysis Platform* (codap.concord.org; Erickson, 2025), a more recent online data analysis tool that builds on the design principles of Fathom and Tinkerplots. It is a cost-free web application in several interface languages. Compared to Fathom and Tinkerplots it can use geographical data and build statistical maps and offers flexible hierarchical data structures. CODAP can be extended by using plug-ins, and we refer to the Arbor plug-in later, when we discuss decision trees. We refer readers to papers by Konold (2007), Biehler et al. (2013), and Finzer and Damelin (2015) for more information about the history of this family of software tools.

Despite this long history, a new focus on "data science education" has led to some new developments in these established tools and technologies that are worth noting here. First, many tools now support or directly integrate more diverse datasets, especially larger, publicly accessible datasets. Tools like CODAP and ESRI's free educational version of ArcGIS Online, a data mapping tool, allow students to easily import a variety of real-world data. New efforts have also made it easier for students to

participate in and access data from citizen science campaigns that are directly relevant to their lives or communities. For example, the Census@School project allows students to contribute to an international dataset recording demographic and behavioral data (Day, 2025), and the WeatherBlur project combined citizen science efforts and environmental probes to allow students to explore weather and wildlife patterns in their local environment (Kermish-Allen et al., 2019). Similarly, some tools, or curricula that are developed to accompany tools, highlight the use of APIs (Application Programming Interfaces) that support students to access dynamic ("live") data from the internet, from sources of data as about topics ranging from weather and sports to music (Fergusson & Wild, 2021; Weintrop & Israel-Fishelson, 2024).

Tools designed for novices are also beginning to highlight new data types and analysis techniques. Some tools and approaches emphasize network data structures and allow students to apply data reasoning to explore traditionally distant domains, such as analyzing relationships described in historical or literary texts (Craig et al., 2021; Legacy et al., 2025). Similarly, curricula and tools have emerged that support students in conducting computational text analysis (Horton et al., 2023) and image analysis. Others allow students to engage with or visualize various predictive modeling and machine learning methods such as decision trees (Biehler & Fleischer, 2021; Erickson & Engel, 2023) and digital classification (Grizioti & Kynigos, 2024). Anna Fergusson (teaching.statistics-is-awesome.org) has developed a suite of tools to help teachers visualize and experiment with simulation-based introductions to a variety of core statistics concepts (Fergusson & Pfannkuch, 2022; Fergusson, 2017).

Finally, as we will review in more detail below, more complex and socially embedded data often demands more interpretive narrative work in order to communicate results clearly and carefully. Many tools for data analysis have started to more explicitly facilitate new ways of communicating the processes and results of a data investigation. Story Builder, a plugin for the CODAP tool, allows students to save "moments" of an analysis cycle and annotate each with photos, rich text, and other explanatory materials (Wilkerson et al., 2021). Similarly, Fleischer and colleagues (Fleischer, Biehler et al., 2022; Fleischer, Hüsing et al., 2022) have begun to explore how students' work with Jupyter Notebooks can be structured to support the construction of "computational essays" that integrate code, outputs (including visualizations), and explanatory text into a coherent narrative. In this approach, half-worked examples and template structures encourage students to write narratives, in the process engaging with issues of data validity and bias, alongside their work through the technical details of an analysis process.

Modern data science practice uses tools with extended libraries that require coding such as R/RStudio or Python. Coding poses obstacles for many, especially younger students. It is an unsolved challenge, how far genuine data analysis practice can be supported at school level without the need of coding, using tools with GUI such as CODAP, and when it is reasonable and possible to move to tools requiring coding and how this transition can be bridged and supported (Fergusson, 2022; McNamara, 2015).

4.3 Critical Data Literacies: New Frameworks for Addressing Socio-scientific and Socio-ecological Problems

In recent years, we can observe two different but related developments, one in statistics and data science education and another in science and mathematical modelling education. In the context of statistics and data science education, we see new emerging frameworks for statistical literacy and data literacy, namely critical data or statistical literacy (Weiland, 2017; Louie, 2022; Martínez-Castro et al., 2023), civic statistical literacy (Ridgway, 2022) or data feminism (d'Igazio & Klein, 2020; Lee et al., 2022). They recognize that data are not neutral but is produced through social processes and

can perpetuate or exacerbate unequal power relations. A further common feature is that the context in which data science should be taught and learned should be actual problems in society and the world, encouraging the development of a critical stance. Similar educational movements include the so-called socio-scientific problem approach in science education (Sadler & Dawson, 2012), socio-ecological problems in mathematics education (Coles, 2023), and the tradition of critical mathematical modeling (Siller et al., 2024; Rosa & Orey, 2015). *Education for sustainable development* and *global citizen education* (which includes, e.g., human rights and democracy education in addition to ecological problems) have also to be mentioned in this context.

In statistics education, an important distinction is between the so-called consumer and producer perspective related to data and statistical products (Gal, 2002). Citizens are much more in the situation of a consumer, when reading newspapers or other documents with data and statistics as compared to doing their own data analysis. Though, from the perspective of education, the experiences and competences gained through conducting one's own analyses may contribute to a critical consumer perspective. The emerging new data literacies are closer to a consumer perspective in the sense that they raise the need to critically reconstruct and deconstruct data uses in science and society instead of inventing completely new ones (Rubel et al., 2021). Advocates of the new literacies sometimes refer to the critical literacy of Freire (2005), who coined the term of *reading and writing the world,* and add "with data" (Weiland, 2022). Writing the world is a metaphor of moving beyond understanding a certain social reality and arguing or working to change it for the better. This of course raises the question of how to deal with political, cultural and social value systems in the classroom.

Data science provides the opportunity to engage students in addressing a range of socio-scientific problems relevant to their lives and communities. This can include exploring environmental issues. Examples include urban planning, participatory mapping, citizen science, and climate investigations (Aridor et al., 2023; Engledowl & Weiland, 2021; Lanouette et al., 2024; Lee et al., 2022; Lim et al., 2023; Louie et al., 2023; Rubel et al., 2021; Wilkerson & Laina, 2018).

For instance, students can analyse some of the "Five Most Important Datasets of Climate Science" (Guimarães et al., 2022), such as CO_2 concentration over the years, using data measured at sites like Mauna Loa, Hawaii (since 1960). For comparison with concentrations before the industrial age, data reconstructed from ice core measurements in places like Vostok, Antarctica (since for over 400.000 years) can be used. The educational challenge is that climate change is not "obvious" from just two data graphs, but evidence relies on theories that explain why measurements taken at two specific places are representative and can be compared at all, and on scientific theories that relate CO_2 concentration to global warming and make plausible that the increase of CO_2 concentration is human-made (Biehler et al., 2025)

Economic issues like the gender pay gap are also fertile ground for critical data investigations. Students can engage in activities exploring the gender pay gap in the U.S., for example, by analysing sample microdata using the new data access tools in CODAP described in Section 4.2 to visualize graphs of wages by sex and educational attainment for a random sample of employed U.S. individuals, using boxplots and dotplots. This allows them to investigate how the gender pay gap varies with the level of education (Louie et al., 2022). On the other hand, the gender pay gap is a complex socio-economic problem, and students will need more background information, for instance what is adjusted and unadjusted pay gap than can be gained through data explorations alone. For instance, the Wikipedia article (https://en.wikipedia.org/wiki/Gender_pay_gap) provides detailed information and several complex graphs that describe historical and geographical variation of the gender pay gap, as well as its variation in different professions, that are usually not taught in statistics

classrooms. This exemplifies the tensions between independent data exploration and critical understanding of argumentative texts of popular science with embedded graphs.

Relegating data to only mathematics and statistics courses may burden such topics, calling for a contribution of the natural and social sciences. Connections between these topics and rich datasets are not hard to find. Many of the topics described above, such as climate and the environment, have a natural home in school science. Data allows students in these courses to also examine deeper intersections between the natural and social, such as through investigations of the disproportionate impacts of climate change on communities of colour and the Global South (Barton et al., in press; Levy & Patz, 2015; Sayavedra et al., under review).

There is a growing literature examining the use and teaching of data within the social sciences (Shreiner, 2018). For example, health issues, prominently highlighted during the COVID-19 pandemic, involve extensive data use and analysis (Calabrese Barton et al., 2021; Engledowl & Weiland, 2021; Rubel et al., 2021). Furthermore, data science is essential for investigating inequalities in society related to race, ethnicity, and migration. Examples include investigating racial bias or uncovering social inequality with data from sources like the U.S. Census (Kahn, 2020; Louie et al., 2022; Rubel et al., 2016).

Different theoretical traditions contribute to shaping the understanding of critical data literacy. In the United States, discussions often center on scrutinizing data for biases and potential misdirection (Engledowl & Weiland, 2021; Kahn et al., 2022), and using data to understand and challenge social oppression and work towards more equitable outcomes (Louie et al., 2022; Rubel et al., 2016). This aligns closely with the principles of teaching mathematics for social justice ("TMSJ"; Gutstein, 2012), which aims to equip students to recognize and counter social injustices using mathematics and data. Key goals include developing social and political consciousness, building agency for social change, and fostering positive identities. This approach views critical data literacy as essential for creating a more just world with data.

The European context, exemplified by initiatives like the ProCivicStat project (Ridgway, 2022), emphasizes Civic Statistics as a means to foster civic engagement through the exploration of evidence. Understanding civic statistics requires comprehending complex social phenomena, often involving dynamic, multivariate data presented through rich texts and novel visualizations. A critical interpretation is essential for informed citizenship in democratic societies, going beyond typical statistics curricula to consider the meaning and implications of data for society and policy. This approach encourages critical thinking and informed decision-making based on evidence. Examples explored in the various chapters include analyzing data related to racial bias in European football, climate change data, or engaging with official statistics and considering their social implications. The project developed a multidimensional framework that shows the multifaceted nature of civic statistics (Figure 3).

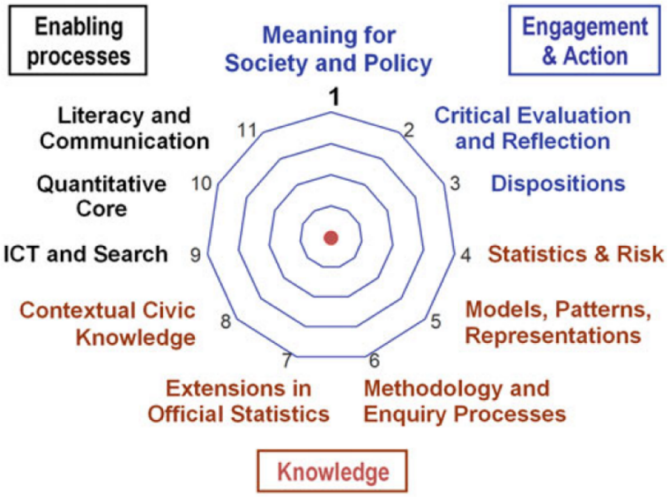


Figure 3. Civic statistical skills, attitudes, knowledge and competences. Graph provided by the authors of Gal et al. (2022).

A particularly rich tradition for understanding critical data literacies emerges from Latin America, heavily influenced by the work of Paulo Freire (2005) and the field of critical mathematics education (Skovsmose, 2014; 2023). This perspective views education as a process of liberation, where individuals develop critical consciousness to understand reality and act to transform it. This involves using statistical investigations to explore societal crises and inequalities, such as repression, conflict, misery, inequity, ecological devastation, and exploitation. The broader field of "critical data studies with Latin America" further elaborates on these ideas, incorporating frameworks of dependence, oppressions, and liberation, viewing datafication as intertwined with existing systems of inequality shaped by class, gender, and race (Campos, 2016; Martínez-Castro et al., 2023; Raffaghelli, 2022; Tygel & Kirsch, 2016; Valente & Grohmann, 2024).

Thus, critical data literacy across these traditions moves beyond technical proficiency to encompass a critical understanding of how data are produced, used, and interpreted within its social, political, and ethical context. It empowers individuals not just to consume data-based arguments critically but to use data actively to understand, challenge, and potentially transform unjust realities. While implementing this approach faces challenges, such as the need for appropriate tools, time, cost, and adequate teacher preparation, the potential benefits for fostering informed and active citizens make it a crucial area for K-12 data science education. However, a central open question is, how to empirically assess the complex goals in knowledge, attitudes, skills and competencies. The available studies do only contribute fragmentary knowledge in specific small-scale studies.

4.4 Presenting the Need for Further Data Literac(ies): From Personal Data Literacy to Datafication Literacy

The datafication of science and society has led to two quite new data-related literacies, which are distinct from what we have discussed so far: *personal data literacies* (Pancrazio & Selwyn, 2019) and *critical datafication literacy* (Sander, 2024a).

Personal data literacies aim at understanding and critically engaging with personal data. This area, sometimes referred to as data awareness, emphasizes that we live in a world where digital devices and platforms are ubiquitous and constantly collect data from us, often without our full awareness. This creates a significant "information asymmetry", where users may not understand where their data

goes, why it is being collected, or how it will be used. The goal of fostering personal data literacies is to help students overcome feelings of powerlessness and develop a sense of self-determination and agency in these interactions.

One framework designed to support this is the *data awareness framework* (Höper & Schulte, 2024; Höper et al., 2024), which provides students with an explanatory model of how data-driven digital artifacts (ddA) work and the role data plays in them. This model helps students become aware of explicit and implicit data collection, the primary and secondary purposes for data processing, and the creation of data models about themselves, often called digital doppelgangers. It also highlights their own role as data sources in these interactions. For instance, one teaching unit designed to build personal data literacy explores location data collected when using a mobile phone, including looking at authentic data like that made available by German politician Malte Spitz. Another unit focuses on recommender systems in streaming services, allowing students to work with a miniaturized engine to see how personalized recommendations are generated based on user data, such as finding "*k* nearest neighbours" to understand how their preferences are used to recommend new films. Learning this explanatory model and applying it to everyday examples seems to encourage students to reflect on these data practices and potentially think about their own behaviour, which is a step toward feeling more empowered rather than resigned (https://www.prodabi.de/en/unterricht/).

Datafication literacy (Sander 2024a; 2024b) broadens the focus from individual data to the larger societal change processes driven by data. In one sentence: The socio-scientific problem addressed by critical datafication theory is the need to critically evaluate the datafication of society and science itself. Critical datafication literacy aims to equip students with the ability to understand and critically reflect on how data are deeply intertwined with social processes and power dynamics An approach that strongly aligns with this is data feminism (d'Ignazio & Klein, 2020), which offers principles such as examining power, challenging power, considering context, and making labour visible, to understand the relationships between data and power. Datafication literacy encourages students to see the labour involved in creating data and to question who benefits and who is marginalized by data practices.

Ultimately, both personal data literacies and datafication literacy are essential for students to navigate our increasingly data-driven world responsibly. Understanding the data collected about oneself (personal data literacies) can serve as a foundational step towards critically reflecting on the broader societal implications of data use (datafication literacy). Issues surrounding the privacy of personal data, for instance, are relevant from both perspectives. However, it's crucial to empower students to look beyond individual actions and understand the need for systemic approaches and collective action to address the challenges posed by datafication. By fostering these literacies, K-12 data science education can move students from passive users to informed and empowered citizens who can critically understand the data that shapes their lives and society and potentially use data to advocate for change.

4.5 AI Literacy from the perspective of mathematics, statistics and computer science education

The data use addressed by these two literacies is also closely connected to data-driven AI systems, particularly those involving machine learning models and algorithms, while also encompassing broader applications of data use that are independent of AI. There has been extensive discussion of AI literacy (e.g. Almatrafi et al., 2024), which analyses these developments in terms of educational requirements.

The emergence of machine learning as a central topic in contemporary AI has led to a paradigm shift in computer science concerning the role of data. Training systems with data instead of classical

programming is the new principle, machines program themselves in a sense, creating algorithms, by being trained with data. This is acknowledged in computer science education, where the new role of data induced a paradigm shift in computer science education itself (Olari & Romeike, 2024; Tedre & Vartiainen, 2023). The topic of what secondary school students should—and are able to—learn about artificial intelligence is primarily explored within the computer science education community. A growing number of publications are addressing this issue, including several comprehensive literature reviews (Almatrafi et al., 2024; Casal-Otero et al., 2023; Chiu & Sanusi, 2024; Olari & Romeike, 2024).

An influential approach are the *five big ideas of artificial intelligence (AI)* of ai4k12.org. They encompass how computers *perceive* the world using sensors, how agents *represent* information and use it for *reasoning* to solve problems, how computers *learn* from data to find patterns (which is machine learning), how intelligent agents interact *naturally* with humans, and the *societal impact* of AI in both positive and negative ways. Data-based machine learning is the closest topic to statistics and mathematics.

Recent literature (Casal-Otero et al., 2023; Long & Magerko, 2020) highlights that AI/ML literacy is not a monolithic concept but spans multiple domains of understanding and engagement. These domains reflect different educational priorities and philosophical orientations toward technology. Three broad categories help to structure current discussions: learning about AI, learning for life with AI, and learning how AI works. The latter entails a conceptual and sometimes technical understanding of how AI systems are developed and deployed. This includes the overall workflow of a machine learning system, from data collection to model deployment, and the role humans play in shaping and overseeing these processes. Within this category, one of the most pressing challenges is addressing what has been termed the multiple opacity problem (Burrell, 2016). Machine learning systems often operate as "black boxes", meaning that the logic behind their outputs is not easily interpretable—even to their creators. This opacity stems from three sources: the complexity of learning algorithms, the abstract nature of the resulting models, and the hidden assumptions embedded in training data.

One approach to dealing with the opacity problem is *black box algorithm auditin*g (Morales-Navarro et al., 2022; 2024). Alternatively, scholars recommend a "white" or "grey" box approach with algorithms where this is possible (Hazzan & Mike, 2023). This means that algorithmic details can be made accessible, at least a simplified version of the professional ones, and reasons for choosing this algorithm. This is very difficult with artificial neural networks, because the required mathematical knowledge about matrices, partial differentiation, and special sigmoid functions are used as activation functions. In contrast, algorithms of methods such as decision trees, k-nearest-neighbours, and linear regression which can be re-interpreted from a machine learning perspective, are mathematically much simpler. We will provide a pedagogical example in section 5.2

4.6 Summary

The frameworks we have reviewed often each come with classroom examples that illustrate the pedagogical approaches and the data science facets that they are designed to support. Although these can be a rich source of guidance for educators, it is clear that not all examples, frameworks, or even data science facets can be taught and learned at K-12 level. Curriculum designers of the various subjects have to select what is suitable and feasible in their educational context. Very recently, the U. S. based *Data Science 4 Everyone* organization, in collaboration with teachers, youth, and experts, have started an initiative to map K-12 data science progressions across the curriculum (DataScienceLearning.org), which is a promising direction though still early in its development.

In the next section, we present pedagogical examples from our own work with colleagues, which of course can illustrate only some of the frameworks and data science facets. These examples serve to illustrate the pedagogical challenges we face already with the relatively small teaching units implemented under very favourable conditions, but also the rich potential such activities hold for conducting truly integrative data investigations even with young students.

5. Pedagogical Examples

To illustrate the many changes and emphasis described above, here we present examples of curriculum focused on data science and data-intensive AI. For each example, we include versions of our curricular approaches that were delivered in excellent conditions, as well as versions that are better adapted to the current constraints of most classrooms and therefore have high potential for immediate scale up. Our goal is to provide some concrete examples of what data science education can look like, and to encourage mathematics educators to consider how such activities might be adopted, supported by, or otherwise intersect with classroom instruction.

5.1 Integrating Data with the Natural and Social Sciences through Data Storytelling in K-12

Teaching data science at the K-12 level presents a significant opportunity to engage students with the world around them. However, moving beyond procedural instruction to foster genuine understanding requires a framework that resonates with how students make sense of complex information. Many approaches have begun to leverage *storytelling* as a potentially powerful, integrative framework for teaching K-12 data science, allowing students to see data analysis not just as a collection of steps or calculations, but as a process of constructing and interpreting meaningful narratives grounded in evidence (e.g., Amato et al., 2022; Finzer & Wilkerson, 2020; Reigh et al., 2022; Sanei et al., 2024; Stornaiuolo, 2020). As we will show, storytelling can make space for students to engage with some of the novel forms of data, workflows, and tools described in Sections 4.1 and 4.2. To the extent that they engage with issues of social and civic relevance, they can also engage students in many of the issues described in Section 4.3, regarding critical data literacies.

In contrast to a traditional statistical report which might review the key features of a dataset, detail the analyses performed, and describe the key results of an analysis and their implications, data storytelling approaches emphasize context and personal expression (Stornaiuolo, 2020), highlighting what might be investigators' motivations for examining a question, providing an accounting of the decisions and processes that underlie the reported analysis (D'Ignazio & Klein, 2020), and invoking aesthetics and emotion to engage the reader and perhaps even persuade them to pursue particular actions as a result of the findings (Reigh et al., 2022). If we consider the producer/consumer distinction introduced above to these ideas about data storytelling, these contextual and social considerations can be written into the stories that students themselves tell as they perform and communicate data analyses, and they can be examined in the stories of others as students work to "read" the stories told in visualizations and statistical reporting in science communication, civic documents, and the media.

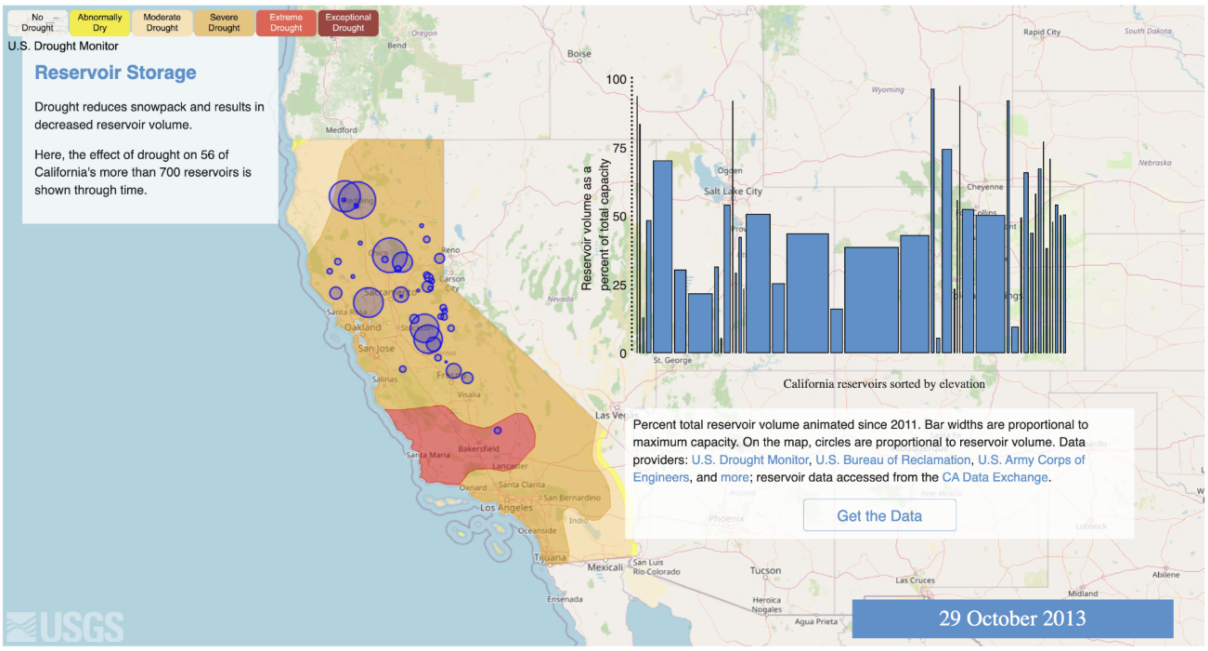
Narrative is often identified as a primary mode of human thought, distinct from but complementary to the logical or paradigmatic thinking often associated with mathematics and science (Bruner, 1991). While logical thinking is associated with decontextualized and verifiable facts, narrative is inherently temporal, contextual, and focused on the meaning and interpretation of events unfolding over time (Healy & Sinclair, 2007; Mor & Noss, 2008). Storytelling provides cognitive and cultural structures to organize thought and motivate coherence in understanding complex situations and multiple perspectives (Bal, 2009; Ochs et al., 1992; Schank & Abelson, 2013). Framing data science through

narrative can bring the analytic and interpretive dimensions of analysis that are so often separated back together (Kemble & Wilkerson, 2024; Pfannkuch et al., 2010).

Example 1: The California Drought. Given the new available tools and technologies described in Section 4.2, students and teachers now have unprecedented access to datasets which often include temporal and geographic features. This creates especially fertile ground for narrative engagement with data. In our work, we have leveraged storytelling approaches to engage science classrooms in week-long investigations of issues at a variety of scales, from personal nutritional decision-making to explorations of regional environmental conditions, to statewide and global climate trends. In these units, a major goal is for students to produce "data stories" that combine data analysis with journalistic research, interviews, personal stories, and expository and persuasive text to highlight issues of personal relevance.

Here, we focus on one investigation we did in collaboration with a 7th grade science teacher, in which students examined water reservoir levels in the U. S. State of California (where this activity was done), from 2011 through 2017. The investigation focused on the question: *What were the main factors that contributed to the California Drought?* We had obtained the dataset and associated visualization that was produced by the U. S. Geological Survey (USGS), and we begin the investigation by exploring a dynamic visualization that was quite complex. It featured a map (Figure 4) that was annotated with circles that changed size to indicate the location and volume of reservoirs and that coloured different regions to indicate regional drought declarations over time. It also featured a bar chart to indicate reservoir volume as a percentage of total capacity. All three of these representations–the reservoir point-locations, regional drought declaration colours, and bar chart–were animated over time. Despite this complexity, the visualization was recommended by the U. S. National Aeronautics and Space Administration (NASA) educational division for use by students as young as 6th grade (students of approximately 11-12 years of age).

There are many patterns, causes, effects, and experiences–some explicit, some implicit, and others only suggested—that a viewer might attribute to the drought as visualized here. When played from beginning to end, it demonstrates dramatic seasonal oscillations in reservoir levels as water is replenished each winter, but also a growing depletion of reserves year-over-year as this replenishment proves insufficient, until the drought finally ends in 2017. Regional patterns demonstrate how the northern regions collect and distribute water to the more populous, and drier, southern cities in the state. Observers with connections to the region may experience emotional reactions as they observe especially dramatic dips in water reserves in 2015, and recall associated public concerns and water use restrictions.



Credit: United States Geological Survey, CC0 1.0 Universal/Public Domain

Figure 4. A dynamic visualization developed by the United States Geological Survey to visualize California drought conditions. As a user scrolls their mouse, the date in the lower right advances and the bar chart and map overlay animate to display changes in reservoir volume across time. We encourage readers to explore the animation at https://bit.ly/cerme14drought or at https://labs.waterdata.usgs.gov/visualizations/ca_drought/index.html.

After exploring this visualization as a class, students were asked to describe what they noticed. We found that students attended to quantitative features inscribed in the visualization, but also to specific locations, time periods, and events represented therein. We also found that many students asked questions about the causes and effects of these patterns and referenced individual water wasting or conserving behaviours, with fewer students also mentioning municipal policies and the impacts of climate change (Barton et al., under review). After they noted these observations, we also provided students with age-appropriate news articles about the drought conditions and impacts. We encouraged students to look across these sources to identify a specific question to investigate related to factors that contributed to the drought, about which they would write a "data story" by combining the provided resources with further data analysis they would conduct throughout the week. Analysing these materials through a storytelling framework encourages students to look for the "plot", and connect resources that might otherwise be treated as isolated.

Based on students' more specific attention and the questions they generated, we then provided the reservoir volume dataset that was used to generate the visualization in CODAP. The dataset included each reservoir's total capacity, location, and the weekly mean volume of water stored for the entire period from Jan 2011 to November 2017. We also made other datasets that addressed students' questions about environmental factors (e.g. monthly precipitation, snowpack, and temperature for each reservoir's nearest weather station) available in CODAP for further investigation. Constructing a data story involves attributing significance and exploring relationships between variables toward a coherent explanation of all factors. As the students engaged with these and other resource materials, competing theories emerged in the class about what is most predictive of drought conditions over time—theories that can be examined, though incompletely, through the lens of data. Students asserted, supported, and analysed one and others' claims using the available datasets, constructing

over time an understanding of the interconnected natural and engineered systems (precipitation, snowpack, water use and distribution via interconnected reservoirs) that impact water access and availability.

This example is one of many from a longer design-based research project exploring students' narrative engagement with data as part of the DataSketch and Writing Data Stories projects (Finzer & Wilkerson, 2020; Kemble & Wilkerson, 2024; Reigh et al., 2022; Roberto et al., 2022; Wilkerson & Laina, 2018). We—Michelle Wilkerson with a team of collaborators including Vasikili Laina, Bill Finzer, Tim Erickson, Hollylynne Lee, Lisette Lopez, Kris Gutiérrez, Emily Reigh, and Cherise McBride—worked in close collaboration with partner grades 7-9 science teachers to identify how curriculum units they already planned to teach could be deepened by asking students to build data stories. Each unit engages students in constructing data stories using a combination of resources including specific publicly available datasets, news stories, scientific argumentation, interviews with family and community members, and personal stories to build connections between data, science, and the social world. It embraces many of the shifts around the pedagogical treatment of data highlighted in Section 4 by leveraging public datasets that incorporate contemporary formats such as GIS and images; interactive artifacts from the media that represent new tools for understanding data in the world; and highlights connections between the data analysed and students' own lives and communities. Additional data storytelling units and associated materials include other dimensions, such as datafication through students' recording and analysis of their own nutritional consumption, can be found at corelab.berkeley.edu and https://bit.ly/wds-units.

Example 2: The DataBytes Discussion Protocol. Weeks-long, open-ended data engagements such as the one described above are not always possible in every classroom. However, there are other ways for students to build deeper interpretive connections to data. One practice that has become increasingly common and can better fit into many existing day-to-day school routines are "data talks", classroom discussions about data visualizations that are intended to build student's everyday competencies as data consumers (Arnold et al., 2022; Boaler et al., 2021; Kahn et al., 2022). Importantly, such discussions are easy to integrate into any school subject, and exploring data visualizations can foreground the relevance of mathematics for understanding topics as varied as global climate, literary sentiment analysis, and consumer opinion.

We developed Data Story Bytes or "DataBytes" to bring a storytelling lens, with attention to the explanatory, social, and contextual features of data analysis and communication, to data talks. DataBytes are a collection of brief structured discussion guides we developed for use with specific visualizations, along with a more general questioning protocol that can help educators lead such discussion about any data visualization (Lee, Wilkerson, et al., 2022; Wilkerson et al., 2025). The DataBytes protocol centres around a four-part questioning framework designed to guide learners through making sense of a data visualization, considering its connections to their own lives and communities, considering the history and context of the data presented, and envisioning how they might use the data and findings in new ways.

To illustrate how the DataBytes protocol works in practice, consider an article from the *New York Times* titled, "Is Sushi 'Healthy'? What About Granola? Where Americans and Nutritionists Disagree" (see https://bit.ly/cerme-nyt; Quealy & Sanger-Katz, 2016). The article features a scatterplot with icons of various food items (e.g. pizza, tofu, coconut oil) positioned to indicate the proportion of "all Americans" (corresponding to the X axis) and the proportion of "nutritionists" (corresponding to the Y axis) that rate each food as "healthy". Already there is interesting mathematical work to be done in the interpretation of this representation, as students are asked to make sense of the position of each food, to reason through the existence and meaning of a dotted line

on the visualization representing $x = y$, and to develop sensibilities around what it means for a food or group of foods to be located "above" or "below" this line.

Moving beyond quantitative interpretation, the discussion guide then prompts students to connect the information and patterns in the data visualization to their own lives and communities. Can they find foods they eat on the graph? Where is their favourite food located, or why might it be missing? Is their own perception of the health of various foods consistent with what is reflected in the visualization? Some might begin to think about how the concept of "healthy" food relates to their own dietary needs, cultural background, or family habits. It is during this phase that we have observed teachers begin to focus more specifically on the scientific context of the data, connecting the quantities indicated in the visualization to the more complex underlying factors and processes they are meant to represent (Wilkerson et al., 2025).

Next, the DataBytes protocol encourages teachers and students to delve deeper into the history and context of the data that are presented within the visualization. This involves considering where the data came from, how it was collected, by whom, and for what purpose (these are called "worry questions" in Gal, 2002)—considerations that might take special precedence as one notices there are particularly salient connections or disconnections between the presentation of data and their relevance to personal or community interests. For example, we find educators and students in this phase begin to question how the specific 52 food items that are featured in the visualization were selected, especially if some of their own favourite foods are missing or appear to be inappropriately assessed. They wonder who was surveyed as part of the "all Americans" or "nutritionists" groups, how those people were contacted, and whether either group may exhibit selection bias that could impact the results.

The final part of the framework asks students to think critically about the data's potential applications, limitations, and future possibilities. This step encourages students to imagine different ways data could be collected or used to answer new questions or represent different perspectives. What would be needed for readers to use the information to make their own dietary decisions? How can the concept of health be indicated more transparently, or represented with more accuracy or precision (for example, through describing macronutritional facts associated with each food, to validate survey participants' responses)? Might the visualization be redesigned to be made more useful or inclusive? In semi-structured clinical interviews with teachers who were considering using the protocol in their own classrooms, we found this step prompted the deepest integration of mathematical considerations with subject matter and social concerns. This is because teachers considered what measurements, calculations, or ways of representing patterns might best reflect the complexities and human impacts of the topic (Wilkerson et al., 2025).

Adopting storytelling as an integrative framework for K-12 data science has helped us to address some of the challenges and necessary adjustments described in Section 4—around the new data types that are now available to explore, the new tools and technologies available, and around social and civic literacies and the impact of data on society. Storytelling is flexible enough to adapt to new representational forms and socially relevant issues. It also can be used across a variety of topic areas and school subjects, allowing for the kind of interdisciplinarity highlighted in the introduction to this chapter. We have found that this approach allows students to engage deeply with data as one part of a bigger story, seeing it as the basis for compelling narratives about the world. Using concrete examples like the California drought, students can explore and model quantitative patterns across time and space while also engaging in the interconnectedness of large, complex systems.

5.2 Integrating Data Science and AI Literacy into Mathematics and Computer Science Education: Decision Trees as an example

We will illustrate the conceptions we characterized in section 4.5. by an example from the ProDaBi project, which is a collaborative project of mathematics and computer science educators, led by Rolf Biehler and Carsten Schulte. Our intention was to integrate mathematical, statistical and computer science aspects into a data science and AI curriculum. We worked with teachers and classrooms in elective courses at secondary level, where no curricular and time constraints limited our experimentation with new ideas and methods, see Biehler et al. (2018) and Heinemann et al. (2018) for the initially different perspectives. One component of the curriculum was machine learning methods. In addition to artificial neural networks, we choose decision trees as relatively transparent machine learning method.

Over the past few years, we—Rolf Biehler, together with Yannik Fleischer and Susanne Podworny—have investigated how decision trees (DT) can be introduced as a machine learning method to students across various grade levels (5/6, 9/10, and 12/13) within the context of the ProDaBi project (Biehler & Fleischer, 2021; Fleischer et al., 2022, 2024; Fleischer & Biehler, 2025; Podworny et al., 2021; 2022; 2025). Classroom materials are available at (https://www.prodabi.de/en/unterricht/), Table 1 provides an overview. The use of DT also illustrates new models of data and workflows with new features (see 4.1). The age verification example we sketch below provides a contribution to critical data literacies (4.3), especially concerning hidden data processing in social media.

Grades and courses	No of lessons	Data	Tools
5/6 (computer science)	10	Nutritional values of food	Data cards (Jupyter Notebook without coding)
9/10 (compulsory elective)	10	Media use	(data cards) CODAP, ProDaBi-JupyterNotebook (without coding)
11-13 (computer science)	16	Media use, data from medicine	Data cards, CODAP, ProDaBi-Jupyter Notebook (without and with coding)
12/13 (elective)	Part of a year-long course	Media use, MNIST data, Parking lot (historical and live data), ..	CODAP, ProDaBi-Jupyter Notebook (without and with coding), coding with Python

Table 1. Decision trees in ProDaBi, different tools and datasets

Exploring decision trees in K-12 education offers a practical way to teach several of the following core concepts, contributing to students'' AI and data literacy. Decision trees are a form of machine learning, directly illustrating how computers can *learn* from data to identify patterns and make predictions. Their transparent, flowchart-like structure makes the hierarchical rule system derived from data relatively easy to grasp, showing students one way knowledge can be *represented* and used for *reasoning* to make classifications or decisions. Furthermore, applying decision tree models to real-world data, such as the media use data of teenagers, clearly demonstrates the *societal impact* of AI, revealing how data collected about individuals can be used to build profiles and influence online experiences through prediction. Discussions around the performance evaluation using metrics like the confusion matrix, and the consequences of different error types (False Positives and False

Negatives) within relatable scenarios (like age verification on platforms), encourage students to think critically about the ethical and societal implications of AI systems that make decisions based on data.

The process of building a decision tree with data cards follows a simplified heuristic of a machine learning algorithm. Students learn to perform a *data split* by selecting a *predictor variable* (like fat content) and a *threshold value* (e.g., more or less than 8 g of fat). The data cards are then divided into two subgroups based on this rule. For each subgroup, the majority label (e.g., most foods in this subgroup are "recommendable") becomes the decision for that branch.

Students evaluate the quality of their decision tree by counting the *number of misclassifications*—the food items that are incorrectly labelled by the rule based on their training data label. To build a multi-level tree, they repeat this process on the subgroups of data that still contain misclassified items, adding more rules based on other predictor variables.

The unplugged approach using data cards offers several benefits. It is a hands-on, active learning experience that helps students deeply understand that a decision tree is a system of rules derived from data. By manually sorting and splitting cards, students can "play the machine", gaining insight into the process before seeing it automated. The relatable context of food makes the topic engaging for young students. Studies on this teaching unit have shown promising results, providing evidence that teaching decision trees as an elementarized data-based model-building process is teachable to middle school students (Fleischer et al., 2024; Fleischer & Biehler, 2025; Podworny et al., 2025)

Decision Trees with CODAP/Arbor in Grades 9-10. After gaining a conceptual understanding through unplugged activities, students can transition to plugged activities using the digital tool CODAP (Common Online Data Analysis Platform, www.codap.concord.com) with the Arbor plugin (see arbor on codap.xyz) for decision trees (Erickson & Engel, 2023). This tool allows students to explore decision trees with larger datasets and witness the automation of the tree-building process. They can also delve into more advanced concepts like evaluating trees with separate test data that was not used for training, understanding overfitting (when a tree performs very well on training data but poorly on test data), and methods like pruning to optimize performance for new data. Combining unplugged and plugged activities provides a balanced approach, where the manual work builds foundational understanding and the digital tools enable exploration of automation and more complex ideas.

The teaching module for grades 9/10 centres around a relevant context for students: analysing data related to teenagers' media use. This is based on the annually conducted representative JIM (Jugend, Information, Medien [Youth, Information, Media]) study in Germany (Rathgeb & Schmid, 2020), which only publishes aggregate data. Our project collected its own microdata with a similar questionnaire, known as JIM-Paderborn (JIM-PB) or YOU-PB, from approximately 1200 teenagers, encompassing 161 questions about their media consumption habits, including digital device ownership, online platform usage (like Instagram, TikTok, YouTube), and gaming. Students are tasked with adopting the role of platform owners using this data to identify users (based on other media use variables) likely to play online games frequently ("frequently" or "rarely") for targeted advertising of a new game. This scenario resonates with students' daily encounters with personalized online experiences.

The Arbor plug-in enables students to build decision trees manually using a visual interface. Students can drag variables from the dataset, which is a reduced version of the full YOU-PB data in this phase for better manageability, into the tree structure. Arbor automatically splits the data based on the variable's values and displays the distribution of the target variable within each resulting branch. Students can assign a predicted label to each branch based on the majority class. A key feature is

Arbor's immediate calculation and display of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) for the current tree configuration. This feedback helps students understand the direct outcomes of their chosen rules, seeing how many cases are correctly or incorrectly classified at each step. By adding more variables in layers, students can manually build and refine a multi-level tree, observing the cumulative effect of rules on classification results.

Arbor supports the comparison of different predictor variables' split in terms of misclassifications, so the students can easily improve their trees. After that, they understand the machine learning algorithms, namely how a full automation can happen if a system tests systematically all available predictor variables and continues to add layers as long as further predictor variables are available.

For our educational purposes, specially prepared menu-based Jupyter Notebooks can make powerful ML algorithms accessible to students without requiring extensive programming knowledge by hiding complex code behind interactive widgets. Yannik Fleischer programmed menu-based Jupyter Notebooks with a simplified version of professional algorithms. Using these notebooks, students can apply a decision tree algorithm to larger datasets, including the full YOU-PB data with nearly 1200 cases and 161 variables. This automated process, where a tree is built in seconds, powerfully illustrates the automation central to ML.

We use a second classification topic with these data, namely pursuing a question. To begin, we posed the question: How might TikTok determine a user's real age based on their behaviour on the platform? This was based on a real-world case from a few years ago, when TikTok sought to identify users under 13—who are not permitted on the platform—by making hidden inferences from their usage patterns. Exploring such implicit inferences is an important aspect of citizenship education in the digital age. Following this introduction, we presented a related and solvable proxy problem using the YOU-PB dataset: How can age be predicted based on other variables, such as patterns of media consumption and ownership of digital devices?" (see Figure 5).

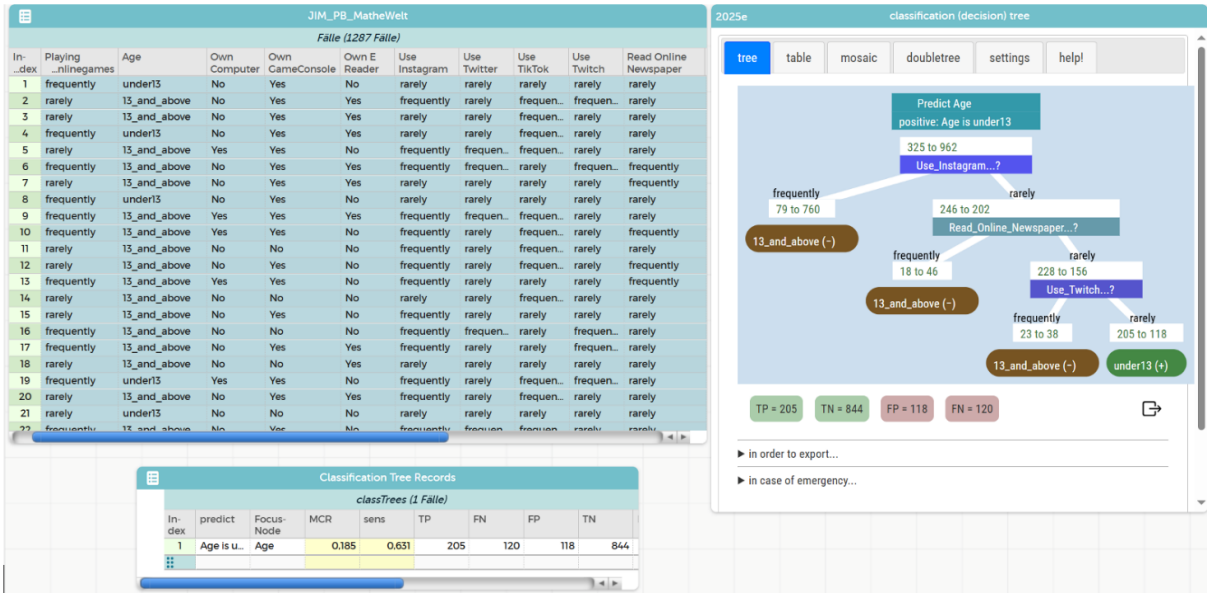


Figure 5. Screenshot from CODAP with plug-in Arbor. The decision tree predicts the variable *age* ("under_13" or "13_and_above") using the variables *Use_Instagram_?*, *Read_Online_Newspaper_?*, *Use_Twich_?* These predictor variables can have the values "rarely" or "frequently". The overall misclassification rate of this tree is 18.5%

The example allowed to discuss the different consequences of two types of misclassifications:

- Predicting that a student is 13 or above 13 years, although younger than 13
- Predicting that a student is younger than 13, although being 13 or above 13 years

The first error is more severe, because the students can use TiKTok although too young. In the second case a ban from the platform is temporary only, because they can present age proofs.

Discussing these nuances is crucial for developing students' critical understanding of how AI systems function and their real-world implications. Plugged tools like Jupyter Notebooks make it straightforward to generate and analyse the confusion matrix and other performance metrics for models built on larger datasets. They also facilitate exploring techniques like pruning, where parts of a potentially overfitted tree are removed to improve its performance on test data. Observing a tree with high training accuracy (e.g., 94.9% in one example) that performs significantly worse on test data (e.g., 75.6%) clearly illustrates overfitting and the value of evaluation and optimization methods.

For students in grades 12 and 13 who are already familiar with Python, the final phase includes hands-on programming: they either develop parts of the algorithm themselves or adapt pre-existing code within Jupyter Notebooks. These notebooks function as worked examples, aligning with the approach outlined by Atkinson et al. (2000). This progression of tools is consistent with the recommendations of Pimentel et al. (2022), commissioned paper for the workshop of the National Academies of Sciences, Engineering, and Medicine (2023), and is supported by research highlighting the educational benefits of using Jupyter Notebooks (Barba et al., 2019) and the instructional value of worked examples (Atkinson et al., 2000).

Further design research is essential to enhance instructional materials, teacher support, and pedagogical strategies. This includes gaining deeper insights into students' prior conceptions, their learning challenges and progress, as well as the specific difficulties teachers face when introducing decision trees in the classroom. While several qualitative studies in experimental classroom settings have already been conducted (Fergusson & Pfannkuch, 2024; Fleischer & Biehler, 2025; Fleischer, Biehler, et al., 2022; Fleischer et al., 2024; Podworny et al., 2025; Zieffler et al., 2021), research aimed at broader implementation and scaling of these teaching approaches remains largely unexplored.

6. Future Challenges: Addressing the multidisciplinary and transdisciplinary nature of data science and AI by staying within and moving beyond discipline-oriented school curricula

At the beginning of this paper, we highlighted Skeina's (2017) assertion that computer scientists "don't respect data" to emphasize the ways in which the domains of mathematics and computer science might be allowing some of the most important features of data—their contextual specificity and social/scientific embeddedness—to slip through the cracks of students' educational experiences. When working with data, Skeina claims, one must "[...] think like a real scientist. Real scientists strive to understand the natural world, which is a complicated and messy place" (p. 2).

It might be no surprise, then, that the examples we present do not actually come from mathematics classrooms, but rather from science and highly contextualized computer science contexts. And even in these cases, we both recognize areas in which our efforts have fallen short. While Michelle has gathered evidence that data storytelling activities engage students deeply in inquiry with data, there is not enough time, space, or preparation in most Science classrooms to address important questions related to validity and uncertainty, such as "how sure are we of this pattern or trend?" More research has begun to focus on the important role of uncertainty in K-12 science instruction (Manz & Suarez,

2018), and this may be one area where deeper data engagements can play a bigger part in the future. And Rolf at times has difficulty convincing collaborators and teachers of the equal importance of situational knowledge and data alongside algorithmic treatment as an important learning goal in Computer Science. Quite a few mathematics teachers are already reluctant to include more data exploration activities in their classroom, given their otherwise full curriculum. Teaching for understanding AI may be assigned to the computer science teacher and ethical, political, and social questions to the social science teacher. However, there is a new trend in mathematics education to explore how generative AI may support teaching and learning processes (Pepin et al., 2025). This may also open the minds for more teaching for understanding the underlying mathematical and statistical principles of AI.

Indeed, effectively integrating data science into the K-12 curriculum presents many challenges. Finzer (2013) calls this the data science education dilemma, lamenting:

> Data science is inherently interdisciplinary, yet it currently has no natural home within the existing, often fragmented landscape of school subjects. Mathematics, while quantitative, often prioritizes abstraction and may be uncomfortable with the rich, messy contexts real data provides. Physical and biological sciences, despite relying heavily on experimental data, are frequently constrained by the need to cover core concepts, pushing data analysis to a secondary, illustrative role. The social sciences, which are increasingly data-driven in practice, have struggled to equip K-12 educators with the quantitative perspective needed for meaningful data use. This lack of a clear disciplinary home means that structured, meaningful opportunities for K-12 students to engage with data analysis are rare, and many teachers may lack the necessary skills and confidence to teach data-driven lessons. (Finzer, 2013, p. 1)

Compounding these practical implementation challenges is the nature of the data science education research field itself, which is also fragmented (Rosenberg & Jones, 2024; Wilkerson, 2025). The field is currently built upon distinct, somewhat isolated clusters of foundational literature that emphasizes data science for undergraduate majors (focused on statistics and computing skills), K-12 education through learning sciences perspectives (focused on data literacy or storytelling), or computational approaches for non-majors (emphasizing specific methods like data mining or simulation). These distinct orientations highlight the need for better communication and collaboration across these scholarly divides.

These orientations are also reflected in the many different definitions and frameworks currently active in the literature, as we briefly reviewed in Section 4.1. We contend that there is nothing problematic in itself about having many different definitions, because these all serve different audiences and purposes. However, these different audiences and purposes should be clearly articulated. The field must stop trying to find a "one size fits all" solution and instead consider what definitions make sense in what contexts, and toward what ends. For example, as described above in Section 4.1, most current frameworks seem most appropriate for the tertiary level and for students who are expected to become specialists in the areas of statistics and data science. And even in our examples, we note that not *all* of the shifts and issues raised in Section 4 are addressed by a single curricular design alone. While they offer examples of what approaching *some* of these new needs and challenges may look like, we are still unsure of *which* challenges are most important to address, and *when*. If we are clearer on these more specific contexts and goals, we can understand how different definitions may build on one another or suggest progressions and paths for students and educators.

Moving forward requires addressing these challenges on multiple fronts (see e.g., Biehler et al., submitted). We need ongoing curriculum development efforts that provide clear progressions of data science concepts and skills, allowing for incremental learning and building upon prior experiences

(see DS4E, 2025 or OECD 2025 for examples). This must be supported by dedicated teacher development and training to build capacity and comfort with data-driven instruction, across subject areas and grade levels (Fergusson & Pfannkuch, 2022; Hudson et al., 2024; NASEM, 2023, Sukol, 2024a; 2024b). Building consensus on standards and core learning goals for K-12 data science is also vital to provide a shared vision and coherence. From a research perspective, fostering deeper engagement and collaboration across existing communities is crucial to build a common literature, shared vision, and convergent understanding of effective pedagogical approaches and learning trajectories.

Ultimately, the goal remains to empower students with the skills and habits of mind necessary to navigate and make sense of the data-rich world. This means incorporating data science thoughtfully across the curriculum, supported by appropriate resources and teacher training. While the field of K-12 data science education research is still coalescing, continued effort to bridge conceptual differences between the disciplines and collaborate on curriculum and pedagogy will be key to realizing the potential of data science education for all students.

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